

$$Pd = 100 - 2u$$

$$\approx 30$$

① Set of rel price, $p = \{p_1, p_2, \dots\}$

o. Accⁿ to Walras law

$$p_1 E_1 + p_2 E_2 + p_3 E_3 + E_n = 0$$

~~1.6~~ As there are only two countries

if $E_i < 0$ $\rightarrow$ excess supply which will work as import for foreign country.

if $E_i > 0$ $\rightarrow$ excess demand export

So for foreign country

$$-p_1 E_1 - p_2 E_2 - \dots - E_n = 0$$

also ~~$p^w M^H = X^H$~~

and $M^H = X^F$, and $X^H = M^F$

and $\frac{1}{p^w} = \text{foreign country}$

$$p^w X^F = M^F \Rightarrow X^F = p^w M^F$$

balanced

② - Property -

③

- (i) Diminishing MP
- (ii) Return to Scale
- (iii) opportunity cost of producing good

→ Decreasing opportunity cost

$$\frac{dx_1}{dx_2} = \frac{L_1^{x_1-1} K_1^{1-x_1}}{L_1^{x_2-1} K_1^{1-x_2}}$$

④ (i) $x_1 \rightarrow CFS$ $x_2 \rightarrow IRS$
 ↳ could be convex

(ii) $x_1 \rightarrow IRS$
 $x_2 \rightarrow IRS$
 ↳ ~~convex~~
 convex

⑤ $MP_{L_1} = A \alpha L_1^{\alpha-1} K_1^{\beta}$
 $= \frac{\alpha}{L_1} x_1$

If $MP_{L_1} > \alpha \cdot \frac{x_1}{L_1}$

$MP_{L_2} = \frac{x_2}{L_2}$

$MP_{K_2} = \frac{x_2}{K_2}$

as $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6 > 0$

∴ If $\alpha_1, \beta, \alpha_2 > 0$
 ↳ increasing MP

also $\alpha + \beta > 1$ $y + d > 1$
 ↳ IRS

① Increasing MP ② IRS

↳ ~~convex~~
 convex PPF

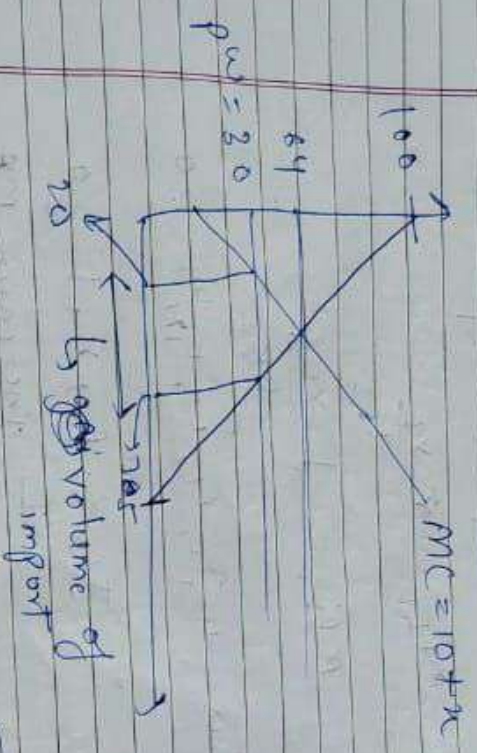
⑦

$TC = 10x + \frac{x^2}{2}$ $MC = 10 + x$

$TR = 100x - 2x^2$

$MR = 100 - 4x$

$MR = MC \rightarrow (100 - 4x) = (10 + x)$



$30 = 100 - 4x \Rightarrow (100 - 4x) = (10 + x)$

$10 + x = 30 \Rightarrow x = 20$

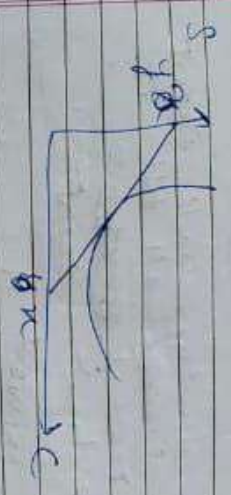
$\rightarrow$ vol of import = 15

⑧ Since cost does not profit max in output level

⑨ Both the goods are produced by CS

Autarky Situation

Linear PPF



$\rightarrow$ Linear PPF

$MKT = \frac{MC_C}{MC_S}$

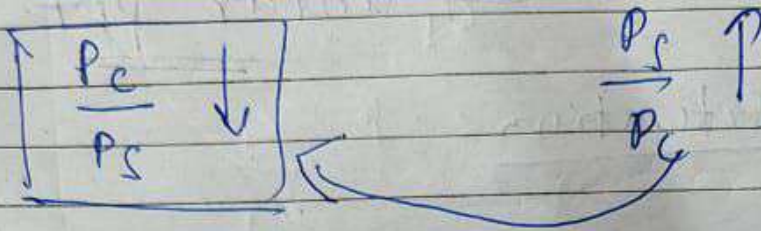
$\frac{P_C}{P_S} = \frac{MC_C}{MC_S} > MC_C$

$\left[\frac{P_C}{P_S} > \frac{MC_C}{MC_S} \right]$

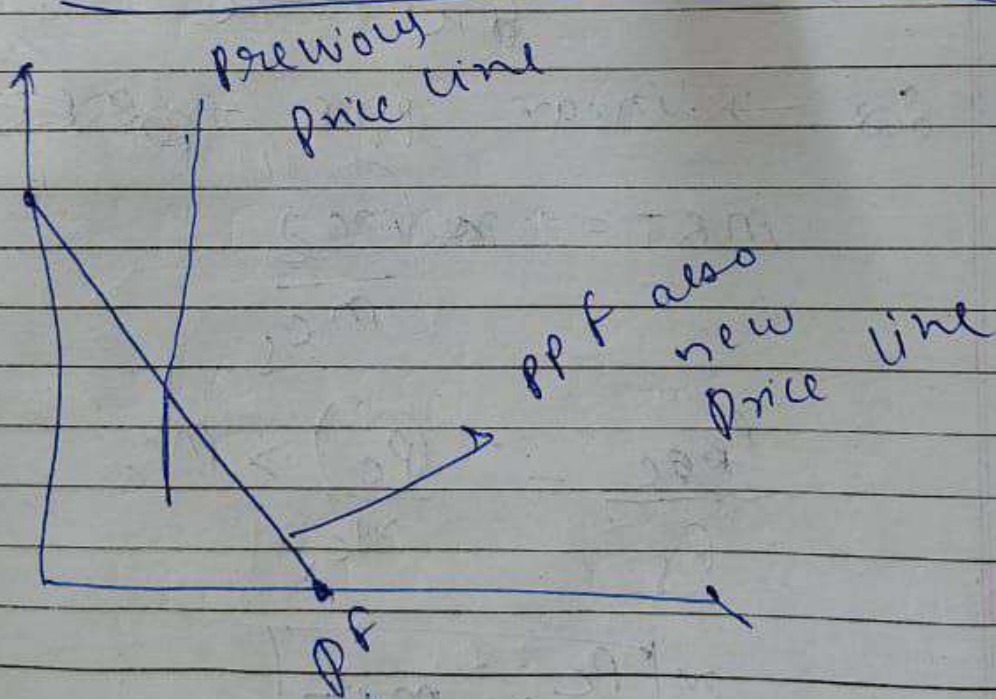


price line is tangency condition is violated

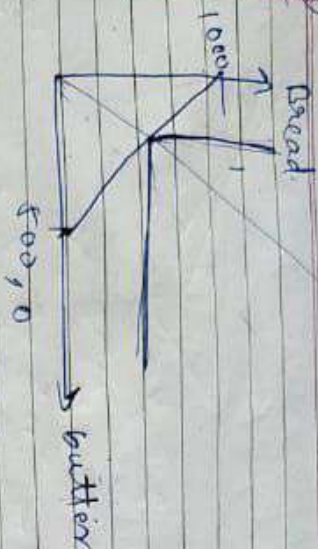
If a country exports S and imports C



So country will export soccer ball till it reaches ~~specialization~~ specialization point



(10)



$$MRT = \frac{500 - 0}{1000 - 0} = -2$$

$$\left[\frac{P_B}{P_M} = -2 \right] \Rightarrow \frac{P_B}{P_M} = -2 \Rightarrow P_B = -2P_M$$

$$\Rightarrow 2 \cdot P_B - P_M = -2(P_M - 500)$$

$$\Rightarrow P_B = -2P_M + 1000$$

$$\Rightarrow P_B = 500 \quad P_M = 250$$

$$P - Y_1 = m(X - Y_1)$$

$$\left[\frac{P_B}{P_A} = -2 \right]$$

(b)

$$MRT_{BM} = -2$$

$$\left[\frac{P_B}{P_M} = -2 \right]$$

(c)

$$\left[\frac{P_B}{P_M} = -0.75 \right]$$

$$\left[\frac{P_B}{P_M} = -0.75 \right]$$

But

$$\frac{P_B}{P_M} < \frac{P_B}{P_M} \Rightarrow P_B < P_M$$

∴ Bread will be exported and Butter will be imported

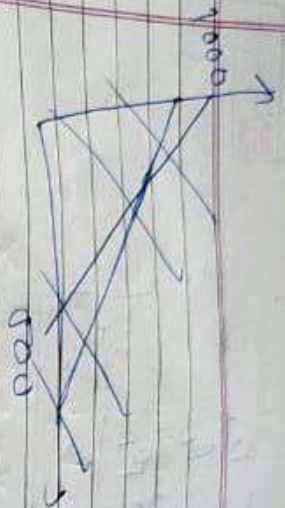
(d)

$$\left[\frac{P_B}{P_A} = \frac{4}{3} \right]$$

→ Produce of Bread

eqn of Price line =

Volume of Bread
export



Vol of export = production - cons at p_u

Production = 750 → complete specialization at p_u

consumption at p_u [B = 2B_u]

$$P_B + P_C + P_{B_u} + P_{B_{u_c}} = 750$$

$$\Rightarrow \frac{3}{4} (2B_u) + B_{u_c} = 750$$

$$\Rightarrow \frac{3}{4} \cdot \frac{10}{4} B_{u_c} = 750$$

$$B_{u_c} = \frac{750 \times 4}{15} = 200$$

$$\rightarrow [B = 600] / [B = 600]$$

Vol of export = $\frac{1000}{1000} - 600 = 400$

(e) (500, 750) → (600, 300)

higher IC
→ Net gain

→ Source of gain specialization effect

→ No exchange gain

(1) $10X_T + X_T X_C = 600 - 20X_C$

⇒ $10X_T = 600 - 20X_C$
 $10 + X_C$

$$\frac{dX_T}{dX_C} = - \left(\frac{-20 \times (10 + X_C) - 600 + 20X_C}{(10 + X_C)^2} \right)$$

$$= \frac{800}{(10 + X_C)^2}$$

$\left[\frac{dX_T}{dX_C} < 0 \right] \rightarrow$ opportunity cost decreasing increasing

pdf is concave

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① $\frac{d}{dx} \left[\frac{1}{1-x} \right] = \frac{1}{(1-x)^2}$

→ $20x_c + 20x_c + 2x_c^2 = 600$

→ $x_c^2 + 20x_c = 300$

→ $x_c (x_c + 20) = 300$

$x_c =$

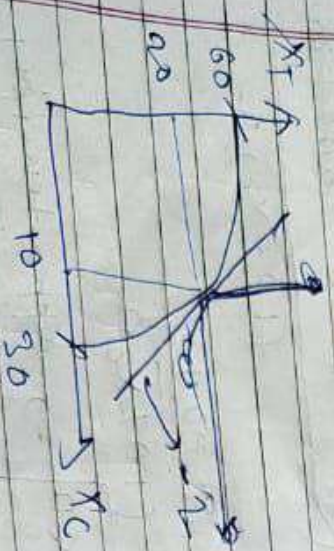
→ $x_c^2 + 20x_c - 300 = 0$

→ $x_c^2 + 30x_c - 10x_c - 300 = 0$

→ $x_c (x_c + 30) - 10(x_c + 30) = 0$

$x_c = 10$ or -30

$x_c = 10$
 $x_T = 20$



②

$\frac{d}{dx} \left[\frac{1}{1-x} \right] = \frac{1}{(1-x)^2}$

→ $10x_T = -1$

$\frac{dx_c}{dx_T} = 2$

$\frac{dx_T}{dx_c} = \frac{1}{2}$

export tonnage



$\frac{800}{(10+x_c)^2} = 0.5$

→ $(10+x_c)^2 = \frac{1600}{0.5}$

→ $(10+x_c)^2 = 3200$

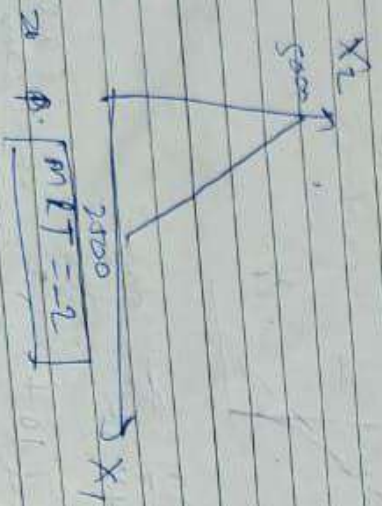
(10 + 12) = 22

$X_1 = 0$

$X_2 = 20$

$X_1 = 0$

(10) $X_1 = 0$



$\frac{P_{X1}}{P_{X2}} = 2$

as

$\frac{P_{X1}}{P_{X2}} = \frac{P_{X1}}{P_{X2}}$

$\frac{P_{X1}}{P_{X2}} = 2$

$\frac{P_{X1}}{P_{X2}} = 1$

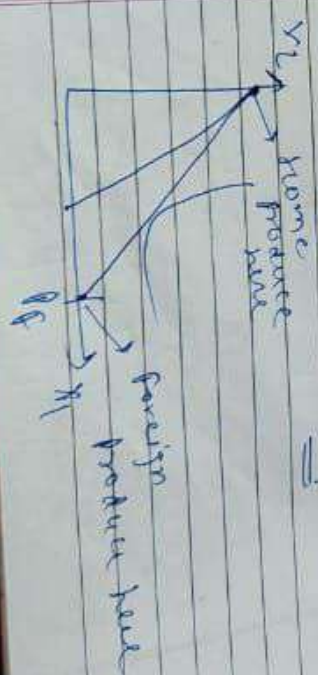
$\frac{P_{X1}}{P_{X2}} = 1$

(10) Same as before

(10) Home country complete specialization in X_2

foreign

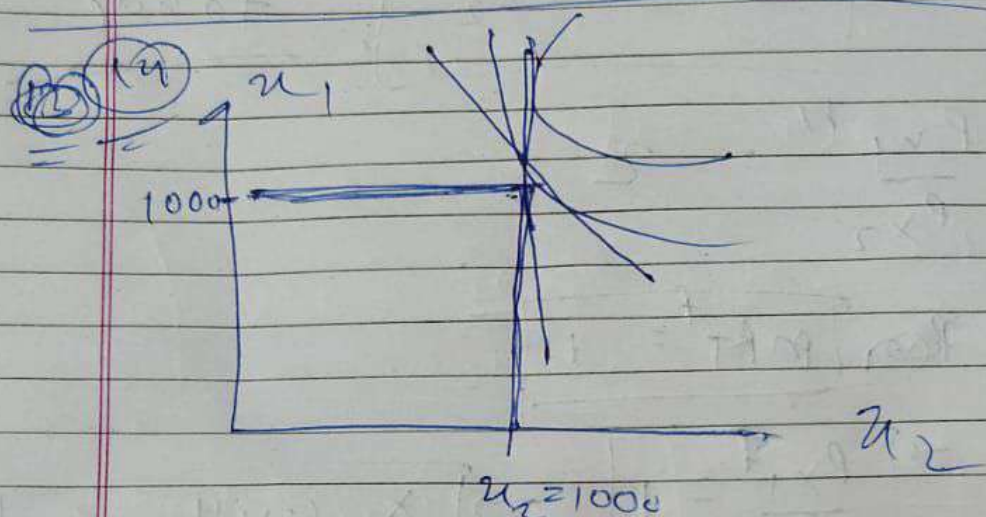
X_1



Net gain for
~~foreign~~ F is zero

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→ product set = $\{(1000, 1000)\}$

→ when inputs are immobile

→ ya there will be gain from
trade through
exchange gain